

You Only Need Clear Images: Self-Supervised Single Image Dehazing

Jiyou Chen , Wenqi Ren , *Member, IEEE*, Huihuang Zhao , Qunbing Xia, and Gaobo Yang 

Abstract—Image hazing refers to adding haze to a clear image, which is important for improving the data amount and diversity of synthetic hazy images that are required to train deep image dehazing models. However, existing image hazing works generate hazy images from a given clear image with a single transmission map. This violates the fact that hazy images are diverse for a natural scene at different times. The domain shift issue between synthetic and real-world hazy images constrains the robustness of deep dehazing models when dealing with real-world hazy images. In this work, we propose an unsupervised haze generation work to synthesize multiple hazy images with diverse haze distributions from a clear image, which requires only an atmospheric scattering model without extra labeling information. Instead of estimating a transmission map from a clear image, we propose to customize the transmission maps by redefining the transmission function. In such a controllable way, hazy images with diverse haze distributions are generated, which avoids the labor-intensive collection of paired data and alleviates the common domain-shift issue of deep image dehazing. Incorporating the unsupervised hazy images generator, we also construct a generalizable self-supervised image dehazing (SSID) framework, where deep image dehazing models can be trained without any human annotations. Extensive experiments on real-world hazy images show that the proposed approach is superior to state-of-the-art unsupervised dehazing works, and achieves competitive performance with the supervised works. Moreover, the proposed SSID framework can be easily generalized to the existing deep dehazing models, greatly improving dehazing robustness on real-world hazy images.

Index Terms—Image hazing, self-supervised learning framework, single image dehazing.

I. INTRODUCTION

HAZE is a common phenomenon, and its presence easily destroys the scene visibility in captured images. Image dehazing refers to restoring a clear image from a hazy one, which is beneficial for successive computer vision tasks such as scene understanding [1] and object detection [2]. To model the haze formation process, a well-known physical Atmospheric Scattering Model (ASM) [3] is defined as

$$I(x) = J(x)t(x) + A(1 - t(x)), \quad (1)$$

where x is pixel coordinate, $I(x)$ is an observed hazy image, $J(x)$ is the clear image to be restored, and A is the atmospheric light. $t(x)$ is the transmission map that models the intensity attenuation as a function of the distance due to light scattering. From Equation. 1, we can know that since the atmospheric light A can be specified manually, hazy image generation is sensitive to the transmission map $t(x)$.

Deep image dehazing models require large-scale datasets for training, which usually learn from paired hazy and clear images in a supervised manner. However, since a scene is either hazy or clear at a specific moment, it is very difficult or even impossible to collect real-world image pairs. Thus, existing deep dehazing models are usually trained on clear images and synthetic hazy images, but they behave poorly when dealing with real-world hazy images. The existing hazy image synthesis works can be divided into the ASM-based and the Generative Adversarial Networks (GANs)-based. The ASM-based works exploit the estimated depth or transmission map to render hazy images. As for the depth map estimation, the existing hazy image synthesis methods either exploit a publicly-released CNN [4], [5], [6], [7], or professional depth scanning devices such as LiDAR and Kinect sensors [8], [9], [10]. They share the similarity that only a single transmission map or depth map is estimated or measured from a given clear image, and the haze distribution between a clear image and its synthesized image is a one-to-one mapping. The GANs-based works treat hazy image synthesis as an image-translation problem. Various GANs were designed to directly learn the one-to-one mapping between a clear image and a hazy image [6], [11], [12]. That is, the existing hazy image synthesis works only achieve a one-to-one mapping for the haze distribution between a clear image and its synthetic hazy image, leading to insufficient diversity of the haze distribution in synthesized hazy images, which easily causes the training of the dehazing model to fall into overfitting. In fact, natural haze distribution is diverse and variable due to complex weather and

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Jiyou Chen and Huihuang Zhao are with the School of Computer Science and Technology, Hengyang Normal University, Hengyang 421008, China (e-mail: jiyouchen@hnu.edu.cn; happyday.huihuang@gmail.com).

Wenqi Ren is with the School of Cyber Science and Technology, Sun Yat-sen University, Shenzhen 518107, China (e-mail: renwq3@mail.sysu.edu.cn).

Qunbing Xia is with Shenzhen Aixiesheng Science and Technology Company Ltd., Shenzhen 518052, China (e-mail: xiaqb@aixiesheng.com).

Gaobo Yang is with the School of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China (e-mail: yang-gaobo@hnu.edu.cn).

The source code is released at <https://github.com/CVhnu/SSID>.
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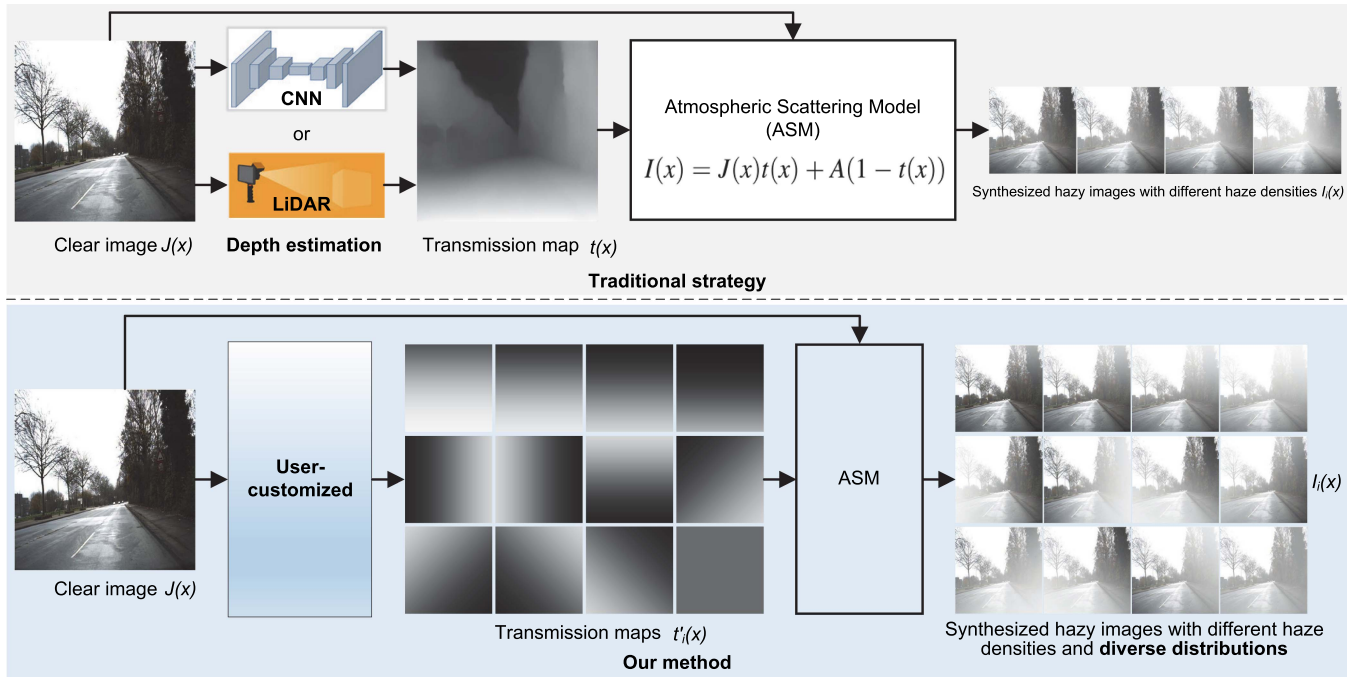


Fig. 1. Comparison between traditional image hazing and the proposed method. The traditional strategy synthesizes hazy images by a single transmission map estimated from CNN or measured from LiDAR, while the proposed method can generate hazy images with distinct haze distributions from a clear image.

environmental conditions. A real-world clear image might have multiple hazy images with different haze distributions captured from the same scene, and there is a one-to-many relationship between them.

As claimed earlier, the existing deep image dehazing works behave well on synthetic hazy images, but they behave poorly when they are generalized to real-world hazy images due to the domain shift issue [13], [14], [15]. To address this issue, some works attempt to enhance the robustness by constructing larger and deeper models, some other works exploit multiple-stage training strategies such as supervised training on synthetic hazy images and unsupervised training on real-world hazy images. They achieve desirable results to some extent but are still unsuitable for practical applications since collecting paired data is labor-intensive and challenging.

In this work, we are devoted to addressing the domain shift issue of deep image dehazing to improve its robustness on real-world hazy images. We propose an unsupervised hazy image generator by redefining the transmission function. Specifically, a concise yet effective transmission function is designed to directly customize diverse, continuous, and depth-aware transmission maps for a clear image. Multiple hazy images with diverse haze distributions can be obtained by exploiting ASM without extra human labels. Moreover, since the parameters of the transmission function can be adjusted to simulate any transmission map, hazy images with any haze distributions can be synthesized from a clear image, thus simulating the one-to-many correspondence between real-world clear images and hazy images. Fig. 1 compares the traditional strategy and the proposed approach. The major difference is the way to generate the transmission map. The traditional strategy obtains a single transmission map from clear images by using CNN or LiDAR, while our work redefines a transmission function to customize diverse

transmission maps in a controllable way. Thus, it can synthesize multiple hazy images with diverse haze distributions but only needs clear images.

Furthermore, we construct a generalizable self-supervised image dehazing (SSID) framework by incorporating the proposed hazy image generator. The framework only requires clear images as input to train the deep image dehazing model, which is optimized by minimizing the losses between recovered clear images and input clear images in a self-supervised manner. That is, the entire deep dehazing model is trained by simply inputting clear images without extra manual labels. Moreover, the existing fully supervised dehazing models can be integrated into the proposed SSID framework without changing network parameters.

Specifically, the contributions of this work are three-fold:

- A novel SSID framework is proposed. Unlike the traditional supervised dehazing works that require paired hazy images and clear ones, this framework only needs to feed clear images to train a deep dehazing model, without any extra manual label.
- To address the domain shift issue that constrains deep image dehazing, we propose to design a re-defined transmission function to customize transmission maps. Hazy images with diverse haze distributions can be synthesized from a clear image in a controllable way, which simulates well the one-to-many correspondence between real-world clear images and hazy images captured from the same scene yet at different times.
- We reformulate the image dehazing task with the generalizable SSID framework. Any deep dehazing model can be integrated into SSID for self-supervised training. This greatly alleviates the requirement of manually labeled training samples and addresses the domain shift issue of deep image dehazing.

II. RELATED WORKS

A. Hazy Image Generation

The existing hazy image generation works can be mainly divided into two classes, namely the ASM-based and the GAN-based.

ASM-based: ASM simulates the formation process of natural haze, and it is exploited by many hazy image synthesis works [4], [5], [7], [9], [10], [16], [17], [18]. Their major differences are the ways to estimate the transmission map. Some works simulate the transmission map by computer software such as SiVICTM [16], [17], some works exploit the publicly-available and well-trained CNNs such as RESIDE-OTS [4], and the others use the depth scanning devices like Kinect or LiDAR [9], [10]. For these works, only a transmission map is estimated or measured from a clear image, leading to a single haze distribution in synthetic hazy images.

GAN-based: GANs have been widely used in image synthesis tasks such as image translation [19]. In recent years, some works [6], [11], [12] have exploited GANs to learn the one-to-one mapping between clear images and hazy images. Zhang et al. [11] proposed a GAN-based density-aware hazy image synthesis work, which learns content features from clear images and style features from hazy images for better rendering of hazy images. Li et al. [12] proposed to treat image hazing and dehazing from the viewpoint of two-way image translation within a weakly supervised framework, which exploits Cycle-GAN to achieve both hazing and dehazing within one network. However, due to the lack of a large-scale image dataset of real-world clear and hazy image pairs for training, the quality of synthetic hazy images is still constrained. Our previous work [6], namely HazeGAN, incorporates the physical ASM and GAN for hazy image synthesis in a coarse-to-fine manner. These GANs-based hazy image synthesis works have achieved impressive results. However, the synthesized hazy images still lack diverse haze distribution.

In addition, some works used software simulation rendering to produce haze effects [17], [20]. They simulate clear images and then render and generate hazy images through graphics algorithms, which present photo-realistic but time-consuming, high-cost, and small-scale images.

B. Single Image Dehazing

Prior-based: They exploit image priors such as dark channel or color attenuation to estimating the transmission map and then exploit ASM for image dehazing [21], [22].

Supervised CNN-based: In recent years, CNN-based works are dominant for image dehazing, which is treated as an image-to-image translation task [23], [24], [25], [26], [27], [28]. Among them, some works introduce GANs into image dehazing [24], [29]. However, it is difficult to collect a large-scale dataset of real-world paired clear/hazy images. As an alternative, these models are usually trained on synthetic hazy images, but they easily lead to the domain shift problem. To address this issue, some works attempt to improve the generalization capabilities of deep dehazing models by semi-supervised learning [13], [14],

[15], [30], [31], [32]. Typically, they are trained on synthetic image datasets in a supervised way to obtain a pre-trained model, and then they are fine-tuned on real-world hazy images in an unsupervised manner. Nevertheless, the pre-trained models are still unreliable due to the inherent limitations of existing synthetic hazy images.

Unsupervised CNN-based: The existing unsupervised dehazing works can be grouped into three categories [33].

The first one only uses a single hazy image or clear image without extra human labels to train the dehazing models. Li et al. [34] proposed a Zero-shot Image Dehazing (ZID) work that uses single hazy images for learning and inference. Later, they proposed another you only look yourself (YOLY) network for dehazing [35]. Due to its layer disentanglement mechanism, YOLY achieves a learning-based haze transfer. However, both models contain multiple complex subnetworks. Especially, there are non-linear and non-concave terms in the disentangle and reconstruction mechanisms, these unsupervised works are difficult to converge when training and the qualities of reconstructed hazy images are unstable.

The second one uses unpaired hazy images and clear images to train the dehazing models, such as USID [36] and UCL-Dehaze [37], etc. They exploit CycleGAN [38] as the backbone, where unpaired natural clear images and hazy ones are used for cycle-consistent adversarial training. USID-Net [36] and D4 [39] are proposed based on CycleGAN for image dehazing, they used cycle-consistency and adversarial loss to constraint the dehazing network trained on unpaired hazy and clear images.

The third one uses paired clear images and their online synthetic hazy images to train the dehazing models [40]. To create the synthetic hazy image via the physical ASM, the depth maps corresponding to clear images are required, which are either provided directly by humans or estimated from a depth network.

Following the definition of the unsupervised paradigm that training models without any human-annotated labels [41], the second and third categories are not fully unsupervised. The second needs unpaired hazy images and clear images, which fall under coarse-grained labels, it is more accurately described as weakly supervised learning. The introduction of depth maps in the third category is also similar, and in this case, the clear image and the synthetic images can only be mapped one-to-one in haze distribution.

Note that the proposed SSID generates hazy images (pseudo labels) by using a designed auxiliary task, which only requires clear images without any extra human labels, so it is a fully unsupervised paradigm. Moreover, the proposed framework is universal, and the supervised dehazing network can be converted into an unsupervised training paradigm by embedding it into the framework. That is, only clear images are required to be fed into the framework without changing any parameters.

C. Self-Supervised Learning

Self-supervised learning is a subset of unsupervised learning, which refers to the learning methods that CNNs are explicitly supervised trained with automatically-generated pseudo

labels from auxiliary tasks without any human annotations [41]. The common auxiliary tasks include image inpainting and image colorization has been introduced for some computer vision works [42], [43], [44].

Compared with the supervised learning that needs paired data X and Y , where Y is the data annotated by human labels, self-supervised learning is trained with X along with its pseudo label P . Note that P is automatically generated for a pretext task without any manual annotation. That is, generating pseudo labels in self-supervised learning is referred to as a pretext task or an auxiliary task. Apparently, the effectiveness of pseudo labels is very important for eliminating manual annotation to achieve self-supervised learning.

Pseudo-labelling is the process of using a labeled data model to predict labels for unlabelled data. Given an image dataset D of K images, $D = \{X_i\}_{i=0}^K$. The pseudo labels $P_i = t(X_i)$ are obtained and fed into the network, where $t(\cdot)$ is the image transformation that generates pseudo labels. The loss function for model training is defined as

$$L_D = \min \frac{1}{N} \sum_{i=1}^K \text{loss}(O_i, X_i), \quad (2)$$

where O_i is the predicted image. The network is optimized by minimizing the similarity between the prediction O and the original input X . Minimizing the loss function forces the network to produce the most similar image representation as its original input X .

III. PROPOSED HAZY IMAGE GENERATOR AND SELF-SUPERVISED IMAGE DEHAZING FRAMEWORK

A. Overview

The proposed hazy image generator is based on customizing the transmission maps. Then, a self-supervised image dehazing framework is established based on the proposed hazy image generator, which further verifies the effectiveness of the proposed hazy image generator to improve the robustness of deep image dehazing when dealing with real-world hazy images.

B. Motivation of Customizing Transmission Map

From Eq. 1, we know that the unknown parameters for image hazing are the atmospheric light A and the transmission map $t(x)$, where A is the atmospheric light that determines the color of haze, and $t(x)$ is the transmission map that models the intensity attenuation as a function of the distance and determines the distribution of the haze. By manually customizing A , multiple hazy images with different haze can be synthesized. As claimed earlier, the existing ASM-based hazy image synthesis works share the common drawback of lacking diverse haze distributions in synthetic hazy images. The inherent reason behind this is that only one transmission map is estimated or measured from a clear image, leading to poor haze distribution diversity in synthesized hazy images.

Different from the ASM-based image dehazing works that require a transmission map as accurately as possible, we argue that image hazing can achieve visually appealing hazy images

with a less accurate transmission map, even if it is not related to the physical structure in a clear image. To evaluate the impact of the accuracy of the transmission map on synthetic hazy images, we blur the transmission map and examine the resultant synthetic hazy images. Fig. 2 shows the synthetic hazy images based on the raw transmission map and blurred transmission maps, respectively. The transmission map 0 is estimated from a clear image using CNN [45], and the transmission maps (1,2,3) are obtained by blurring the transmission map 0. We can observe that though there are great differences among the transmission maps (0,1,2,3), the perceived differences of synthesis hazy images (0,1,2,3) are slight. Note that the physical structure in the clear image is almost lost in the transmission map 2, but the corresponding synthetic hazy image 2 is still photo-realistic, which is visually similar to the synthetic hazy image 0. This motivates us that it is possible to customize transmission maps directly without referring to clear images for hazy image synthesis.

We randomly select two clear images, and their transmission maps are obtained by LiDAR and CNN, respectively. We also customize two similar transmission maps without referring to any information from the clear images. Finally, ASM is used to synthesize four hazy images with four different transmission maps. Fig. 3 compares the synthetic hazy images from the same clear one. We can observe that though our customized transmission maps lose the physical structure from the clear image, the synthetic hazy images have comparable or even better visual qualities, when compared with the other hazy images by exploiting the transmission maps by LiDAR and CNN. This proves that without any physical structure information of a clear image, it is still possible to achieve high-quality hazy image synthesis. Therefore, we propose to customize four transmission maps with gradual changes for a clear image, and then use them to synthesize hazy images with diverse haze distributions. In addition, to synthesize non-homogeneous haze, we rotate the transmission map to generate hazy images with distinctive haze distributions. Fig. 4 shows four synthetic hazy images based on the raw depth map and three rotated depth maps, respectively. From it, we can see that transmission depth maps were obtained by rotating the raw transmission map of 90° , 180° , and 270° , and their synthetic hazy images look photo-realistic, which simulate well natural hazy images with diverse haze distributions. In theory, it is possible to obtain multiple depth maps by rotating the raw one at any angle, to generate more synthetic hazy images with diverse haze distributions, thereby simulating the one-to-many relationship between clear images and natural hazy images. In the following, we propose to customize the transmission maps in a controlled manner by redefining the transmission function.

C. Redefining of Transmission Function

The transmission function is redefined based on the following observations: natural haze distribution, including trend, direction, and density, is easily affected by weather. Moreover, natural haze is usually continuous. As a result, the haze distribution should also be continuous in synthetic hazy images. From the ASM defined in Equation 1, A , $J(x)$, and $I(x)$ are coplanar

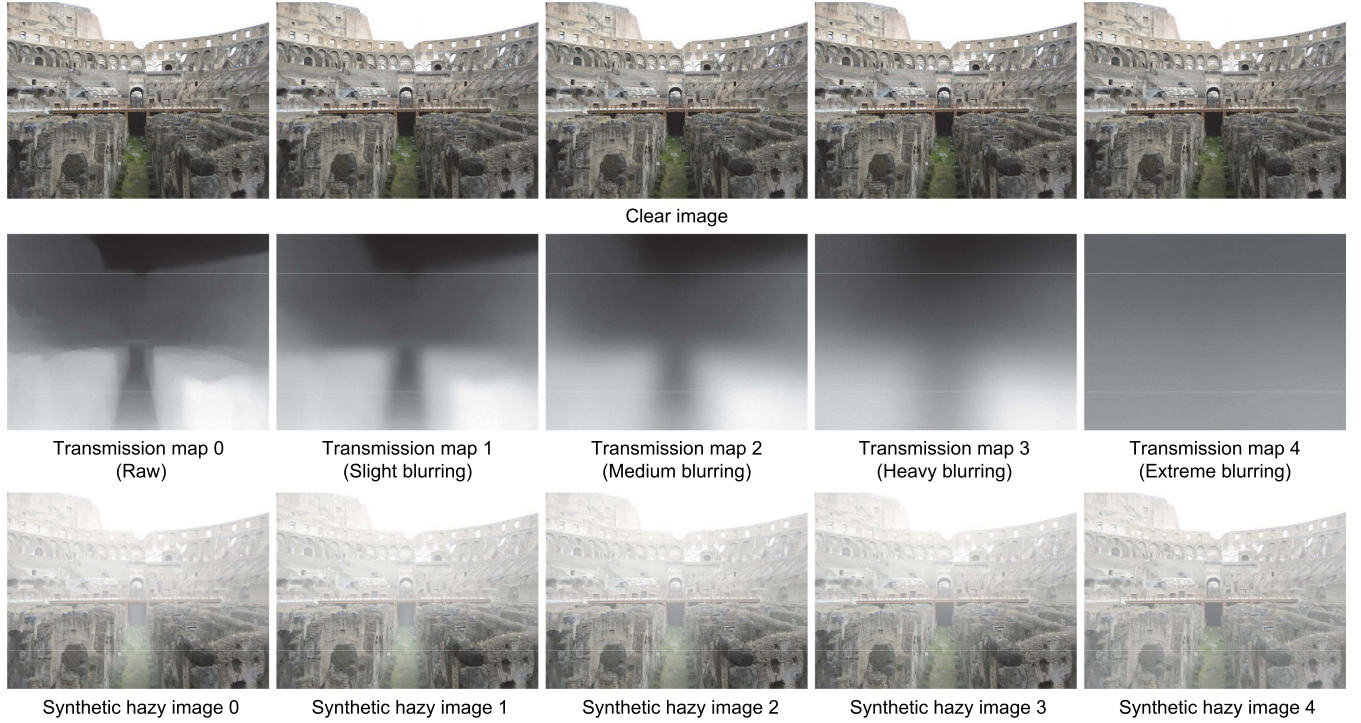


Fig. 2. Example synthetic hazy image generated from the same clear image and different depth maps via ASM. Three different degrees of blurring are processed in the raw depth map, and then the corresponding hazy images are synthesized, respectively. The synthesized hazy images show that synthesizing hazy images may not need an accurate transmission map.

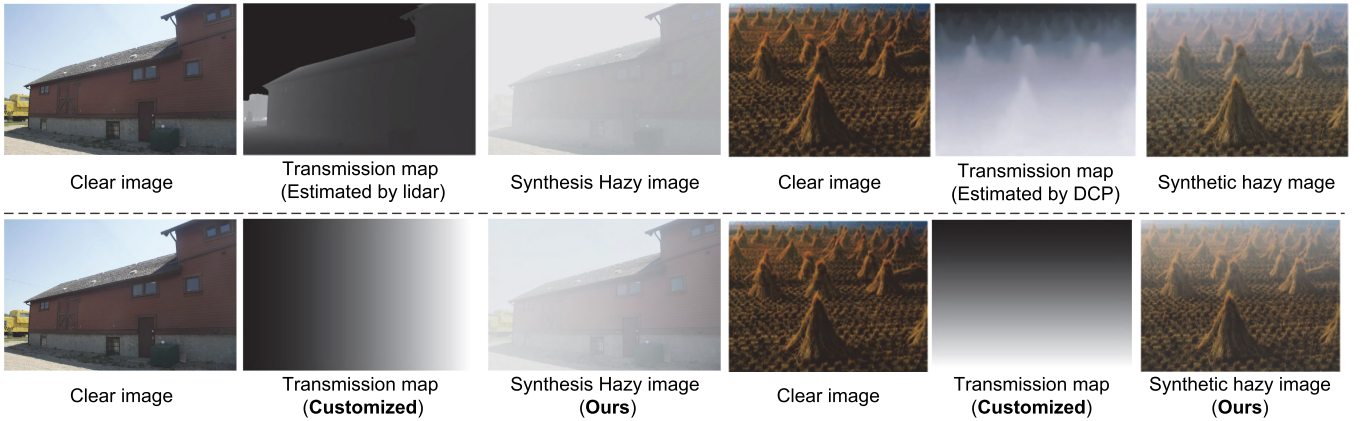


Fig. 3. Example of the hazy image synthesized from transmission maps obtained by LiDAR, dark channel-prior, and our customized, respectively. The quality of our generated hazy images is equivalent to or even better than the others, and customized transmission maps are easy and readily available.

and their endpoints are collinear. Thus, the transmission $t(x)$ is the ratio of two line segments:

$$t(x) = \frac{\|A - I(x)\|}{\|A - J(x)\|} = \frac{A^c - I^c(x)}{A^c - J^c(x)}, \quad (3)$$

where $c \in r, b, g$ represents the color channel.

From Equation 3, we know that if the atmospheric light A is changed, there will exist corresponding transmission maps $t(x)$. As shown in Fig. 5, for a clear image $J(x)$, when the atmospheric light A changes, the ratio of two line segments will also change, producing multiple different transmission maps $t(x)$. This implies that there is a one-to-many relationship between

clear images $J(x)$ and transmission maps $t(x)$. Thus, it is theoretically feasible to synthesize diverse hazy images from a clear image by designing multiple transmission maps.

Traditional, the transmission map $t(x)$ can be expressed by Koschmieder's law [3]:

$$t(x) = e^{-\beta d(x)}, \quad (4)$$

where $d(x)$ is the depth between an object in the image and the observer, which can be obtained by the depth estimation algorithm, and determines the shape distribution of the haze. β is the attenuation coefficient controlling the thickness of the haze. Although multiple hazy images with different haze densities can

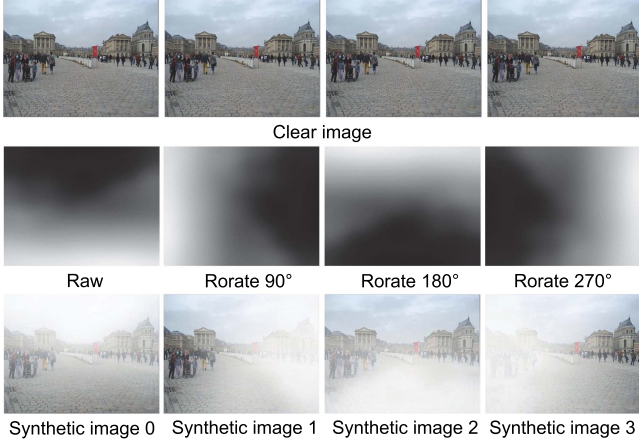


Fig. 4. Example synthetic hazy image generated from the same clear image and different transmission maps. Three transmission maps were generated from the raw transmission by rotating 90°, 180°, and 270°, respectively.

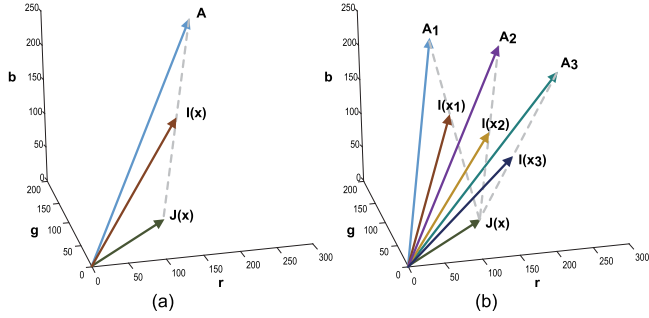


Fig. 5. (a) Image hazing model. (b) Multiple hazy images $I_i(x)$ synthesized with different atmospheric light A_i from a single clear image $J(x)$.

be synthesized by changing β , their shape distribution of haze is single due to the uniqueness of $d(x)$, which is inconsistent with the diversity of haze in nature.

Based on the above, we know that the existing transmission function (4) has only one parameter β to control the haze density of synthetic hazy images, resulting in one-to-one mapping for the haze distribution between a clear image and its synthetic hazy images. However, natural haze distribution is diverse in terms of both haze density and distribution, and there is a one-to-many mapping between a clear image and its hazy images. Thus, hazy image synthesis should consider haze density and distribution simultaneously, which need two parameters to control them. The cumulative distribution function of Weibull distribution [46], which has a shape parameter and a scale parameter to control the distribution characteristics. Inspired by this, to generate hazy images with diverse haze density and distribution, we follow the Weibull distribution to redefine the transmission function as follows:

$$t'(x) = \frac{1}{1 + e^{-k(x-m)}} d'(x) + b. \quad (5)$$

where $d'(x)$ is a pseudo depth map with the values initialized to 1 and the same size as the clear image, x is the normalized pixel coordinate, k is the coefficient related to the shape of the curve,

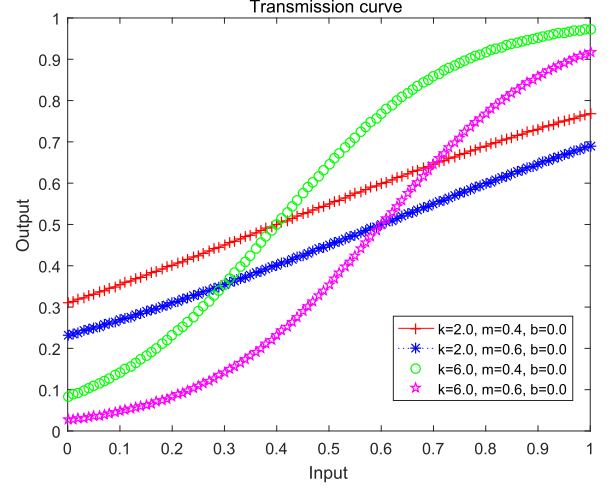


Fig. 6. Examples of the transmission curves, where different curves indicate the different mapping relationships between the input depth map and the output transmission map.

which mainly controls the density of haze, m is the coefficient is related to the scale of the curve, which mainly controls the distribution of haze, and b is the bias coefficient to prevent pixel values from crossing the boundary, and the default value is 0. Based on this, the ASM is redefined as

$$I(x) = J(x)t'(x) + A(1 - t'(x)), \quad (6)$$

where $I(x)$ is the synthesized hazy image, $J(x)$ is the clear image, and A is the atmospheric light.

Compared with the native transmission function 4, the redefined transmission function is more flexible. It alleviates the limitation that the depth map is closely related to a clear image. Adjusting the above three parameters k , m , and b can synthesize a hazy image with any haze distribution. Fig. 6 shows the mapping curves of transmission, different k , m , and b have different curves, and each curve represents a unique mapping relationship between the input and the output, which also represents that a unique haze distributed hazy image can be synthesized. Fig. 7 is the process of customizing the transmission maps and generating the hazy images. We can observe that compared to the synthesized hazy image with the parameters $k = 2$, $m = 0.4$, $b = 0$, the synthesized hazy one with parameters $k = 6$, $m = 0.4$, $b = 0$ has thicker haze. Similarly, compared to the synthesized hazy image with the parameters $k = 6$, $m = 0.4$, $b = 0$, the synthesized hazy one with parameters $k = 6$, $m = 0.6$, $b = 0$ has a different distribution. These diverse synthetic hazy images demonstrate the effectiveness of the proposed redefined transmission function, and they appear very realistic and exhibit diverse haze distributions.

D. Pipeline of Image Hazing

First, we define a depth map with the same size as the input clear image, and all the elements are initialized to 1.

Second, a redefined transmission function is designed to adjust the matrix value non-linearly, and multiple directions are computed to obtain multiple transmission maps. Note that the

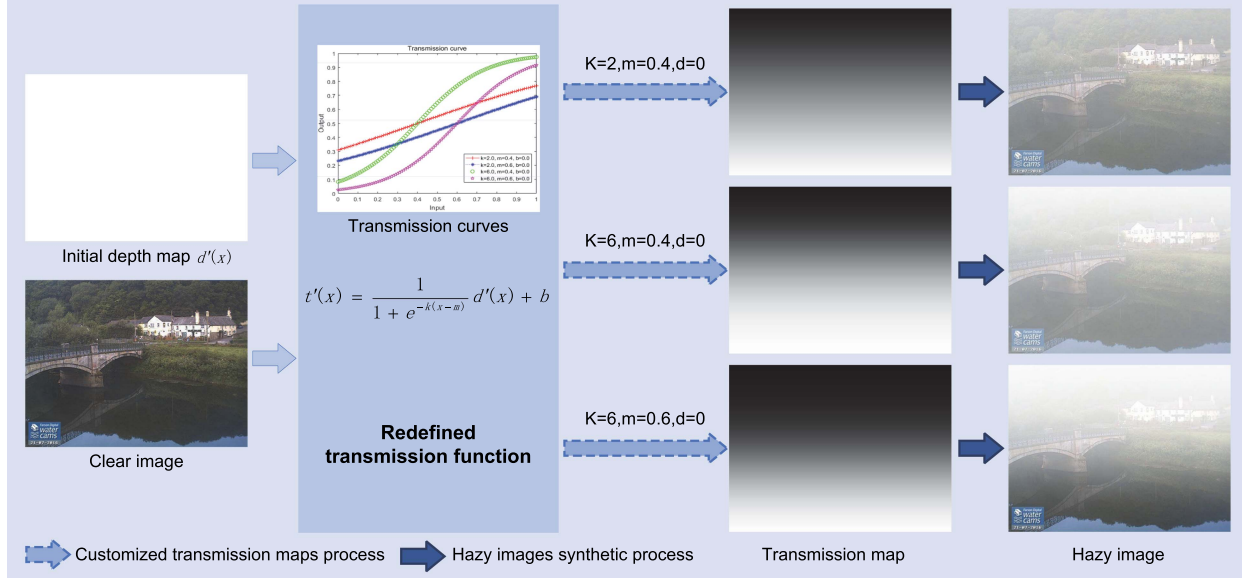


Fig. 7. Illustrate the process of redefined transmission map and hazy image generation.

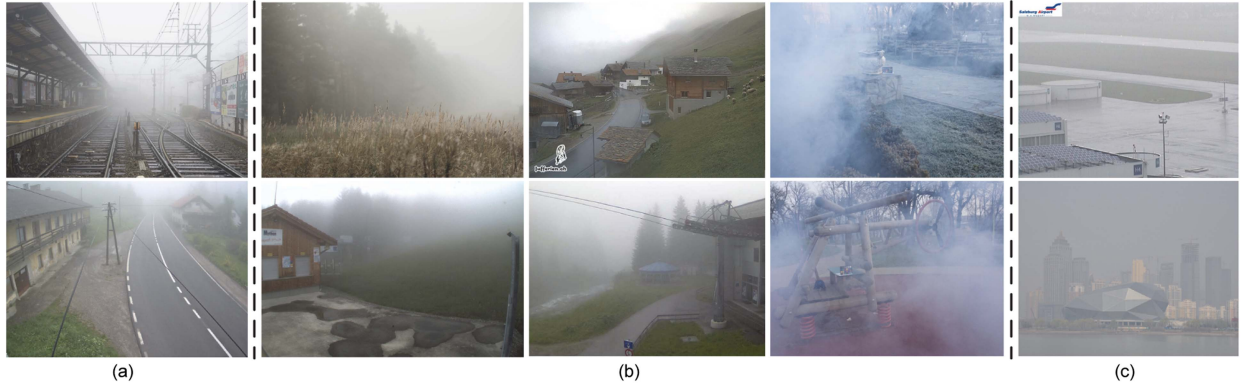


Fig. 8. Illustrate the diverse haze distributions of real-world hazy images. (a) the haze density increases with scene depth. (b) non-homogeneous haze is distributed in local areas. (c) homogeneous haze distribution.

direction here refers to the changing trend of pixels in the transmission map, and different trends lead to more diverse haze distributions. Generally, humans perceive that the haze density increases with scene depth. However, the distribution of natural haze is closely related to multiple factors such as temperature, humidity, wind direction, etc., which leads to the diverse distribution of haze, such as local, clustered, and uniform. Fig. 8 shows different haze distribution images in the real world. We can observe that some are depth-dependent and uniformly distributed, especially the non-homogeneous haze distribution in local regions (Fig. 8(b)). Thus, we define multiple directions of the transmission maps to simulate natural diverse haze distributions.

Third, to simulate various haze distributions in the real world, eight transmission maps with different directions are defined. let $t'_1(x) \sim t'_8(x)$ be eight transmission maps. All direction-specific quantities are denoted by the superscripts $\{\leftarrow, \rightarrow, \uparrow, \downarrow, \nearrow, \searrow, \swarrow, \nwarrow\}$, which reflect the directions for calculation. Theoretically, any direction haze distributions can be synthesized by Equation 5. Finally, the customized transmission

map $t'(x)$, the customized atmospheric light A , and the input clear image $J(x)$ are fed into Equation 1 for hazy image synthesis.

Fig. 9 shows the pipeline of image hazing. By customizing the transmission function, nine distinct transmission maps are obtained, representing different haze directions and shapes. Then, these maps are applied to synthesize nine hazy images with diverse haze distributions. In fact, by controlling the parameters in the redefined functions, more hazy images with different distributions can be synthesized.

E. Self-Supervised Dehazing Framework

To learn the visual representation of an unlabeled image in a self-supervised manner, the conventional way is to design an auxiliary task, which automatically generates high-quality pseudo labels. For image dehazing, designing the auxiliary task should meet the following two requirements. Firstly, hazy images are automatically generated without any human annotation involved. Secondly, pseudo labels, namely the generated hazy

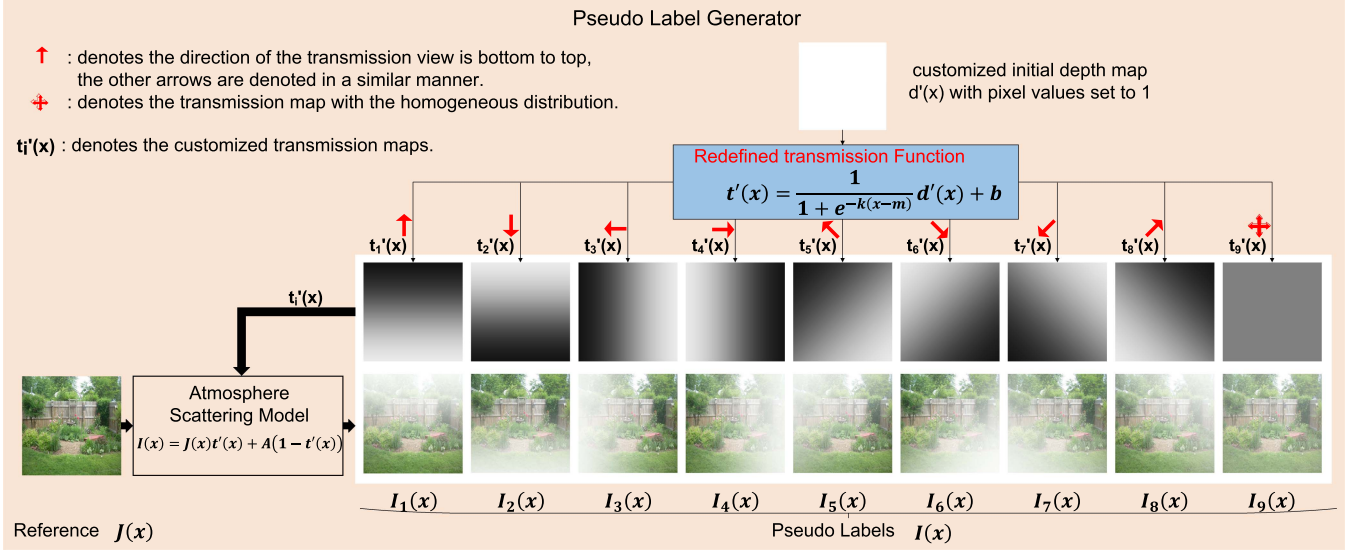


Fig. 9. Illustrate the details of the hazy image Generator. Eight directions are inhomogeneous and one homogeneous haze distribution to make the synthesized hazy images with diverse haze distribution.

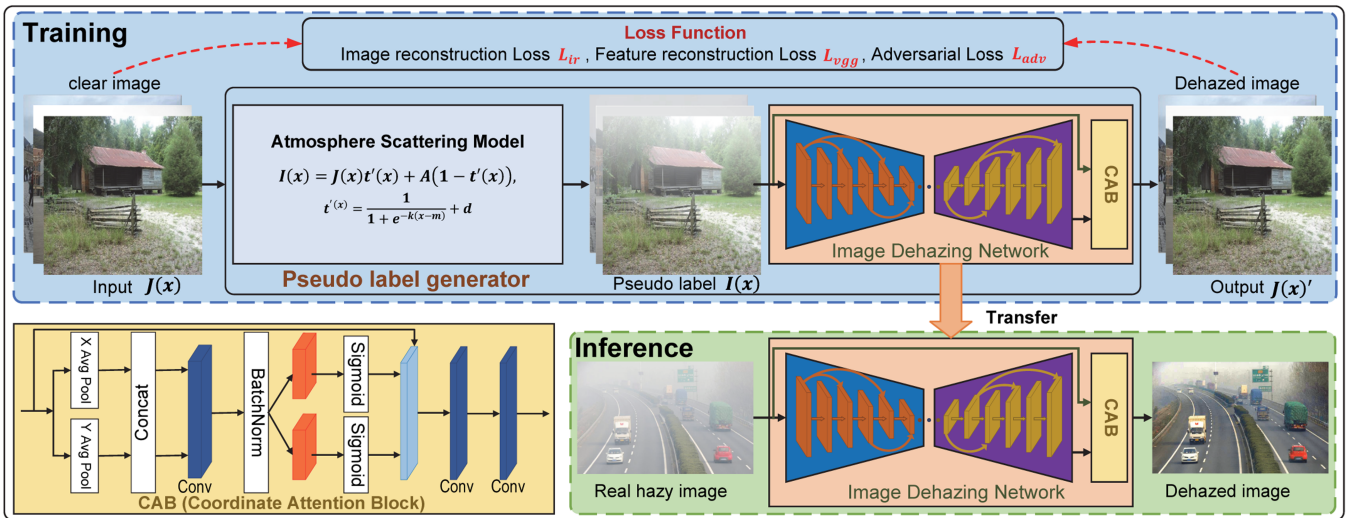


Fig. 10. Illustrate our self-supervised learning image dehazing framework. Different from the supervised training that requires feeding pairs of hazy images and clear references, our self-supervised framework is like a “black box”, and users only need to input clear images to train the dehazing model.

images, should be as photo-realistic as possible and be diverse in terms of haze distribution.

Based on the proposed hazy image generator, we construct a self-supervised image dehazing (SSID) framework. As shown in Fig. 10, which is made up of a hazy image generator and a deep dehazing network. Input clear images are fed into the hazy image generator to generate synthetic hazy images, which are input into the single image dehazing network to restore clear images. That is, the framework is trained in an end-to-end manner, which needs only the input of clear images. To prove the effectiveness of the proposed framework, we simply adopt ResNet18 [47] and coordinate attention [48] to construct a deep dehazing network, which is made up of four modules including encoder, decoder, attention, and discriminator.

The workflow of the self-supervised learning-based image dehazing framework is divided into two stages, namely training and inference. In the training stage, a clear image is input to the hazy image generator to synthesize the corresponding hazy images. Then, the synthesized hazy images are fed into the image dehazing network to obtain the restored images. Finally, the restored images and the original input clear image are used to calculate the loss, which is used to update the weights of the image dehazing network. The training of the image dehazing network is stopped when the restored image is close to the input clear image, saving the parameters of the image dehazing network. The training is end-to-end, and the inputs are only clear images and without extra manual labels. In the inference stage, a real hazy image is fed into the

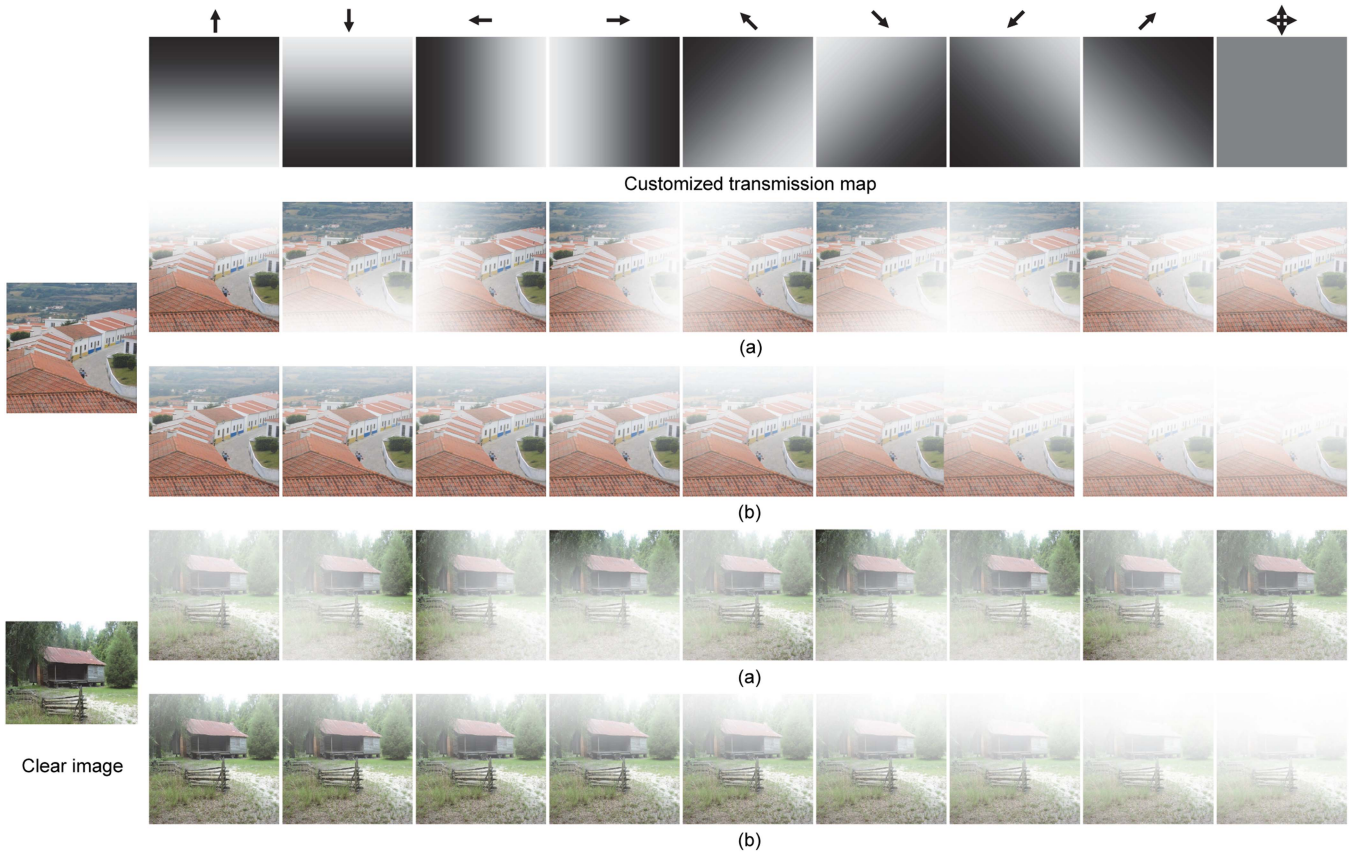


Fig. 11. Illustrate the synthetic hazy images by our pseudo label generator. (a) Multi-distribution results are synthesized by controlling the parameter k . (b) Multi-density results are synthesized by controlling the parameter m .

well-trained image dehazing network, which outputs a dehazed image.

IV. EXPERIMENTS

In this section, we compare synthetic hazy images by the proposed hazy image generator and the existing works. Then, we compare the self-supervised image dehazing network with the SOTA image dehazing methods.

A. Experiment Settings

Train datasets: Our image dehazing model only requires clear images for training. As we all know, the weather before haze occurs is usually cloudy. To make the synthetic hazy images much closer to the ones captured naturally, we first carefully select 5,000 clear images in cloudy from three datasets including RESIDE-OTS [4], RSCM [49] and TCWC [50], and then exploit the proposed hazy image generator to obtain synthetic hazy images as the pseudo label. Fig. 11 shows some example hazy images by the proposed hazy image generator. To make fair comparisons, we directly use the open-source codes of the baselines. In addition, all compared models are trained with the same number of image pairs.

Test datasets: To evaluate the robustness of the proposed SSID framework when dealing with a wide range of real-world scenarios, it is tested on four real-world datasets. RTTS [4]

is a real-world hazy image dataset widely used for evaluating deep dehazing models. RW-HAZE [51] is a real-world image dataset of well-paired clear/hazy images from a weather monitoring station for full-reference dehazed image quality assessment. I-HAZE [52] and O-HAZE [53] are real-world hazy image datasets for image dehazing evaluations. The difference from RW-HAZE is that a professional haze machine simulates that haze, and then the camera captures pairs of clear and hazy images.

Competitors and Metrics: Two large-scale synthetic hazy image datasets RESIDE [4] and Foggy Cityscapes [10] are used for making comparisons with the synthetic hazy images by the proposed hazy image generator. In addition, the proposed image dehazing framework is compared with state-of-the-art (SOTA) unsupervised image dehazing works, which include Deep DCP [54], RefinedNet [55], YOLY [35], USID [36], D4 [39] and UCL-Dehaze [37]. The metrics include five specifically designed for image dehazing quality metrics, including four full-reference metrics such as DehIQA [56], FRFSIM [57], VI [58], RI [58], and one no-reference FADE [59].

Training strategy and parameters: The proposed framework is implemented on PyTorch 1.3.0 and Python 3.6.9 on an Intel(R) Core(TM) i7-7800X CPU @ 3.50GHz, 32GB RAM, and two Nvidia RTX 3090 TI GPUs. We adopt an Adam optimizer with the exponential decay rates of $\beta_1 = 0.9$ and $\beta_2 = 0.999$ to train our model, and the batch size is set to 16. The learning rate is

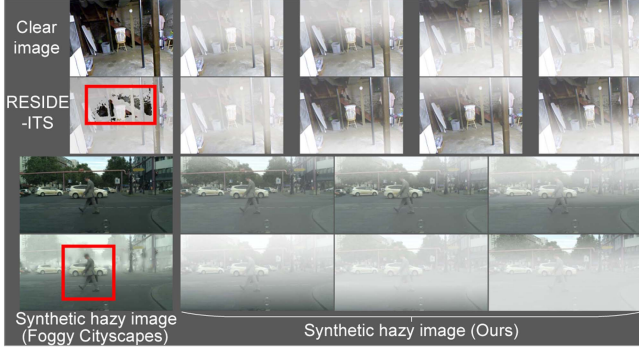


Fig. 12. Illustrate the hazy images from RESIDE [4], Foggy Cityscapes [10], and our generated hazy images with different distributions.

set to 1×10^{-4} and is multiplied by 0.1 after 10 epochs. Three parameters k, d, m are empirically set in Eq. 5, k is in $[2.0, 6.0]$, m is in $[0.2, 1.0]$, and b is in $[0.00, 0.10]$ to generate diverse transmission maps. In addition, A is random in $[0.6, 1.0]$ in Eq. 1 to ensure that the synthetic hazy images have different global atmospheric light.

Training loss: Three of the loss functions including image reconstruction loss, adversarial loss, and perceptual loss are applied to feedback and update the parameters of the dehazing network, until the loss between the dehazed image and the reference one is no longer decreasing in five epochs. The input clear image is fed into the “black box” consisting of the pseudo label generator and image dehazing network to generate the dehazed image. Then, the above three losses are calculated between the dehazed image and the input clear one (Ground truth). Specifically, the image reconstruction loss L_{ir} , adversarial loss L_{adv} , and perceptual loss L_{vgg} are defined as follows:

$$L_{ir} = \|y - G(x)\|, \quad (7)$$

$$L_{adv} = E_{(y)} \left[(D(y) - 1)^2 \right] + E_{G(x)} \left[(D(G(x)))^2 \right], \quad (8)$$

$$L_{vgg} = \frac{1}{C_i H_i W_i} \|\phi_i(G(x)) - \phi_i(y)\|_2^2, \quad (9)$$

where y and $G(x)$ denote the clear images and the dehazed images, respectively. C_i denotes the image channel, $\phi_i(\cdot)$ is the activation of the i -th layers of VGGNet, H_i and W_i are the height and width of the i -th feature map, respectively. The final loss function is defined as

$$L_{SSID} = \lambda_1 L_{ir} + \lambda_2 L_{adv} + \lambda_3 L_{vgg}. \quad (10)$$

λ_1, λ_2 , and λ_3 are set to 5, 1, and 5, respectively.

B. Comparison With Benchmark Hazy Images

Figs. 12 and 13 show the qualitative comparison results of our generated hazy images with the existing synthetic and the real-world ones, respectively. Specifically, due to the inaccurate depths estimated by CNN, the hazy images from RESIDE have some artifacts near the edge of objects, and the ones from Cityscapes look inharmonious at the pedestrian, which is marked with red boxes. In contrast, our hazy images look more

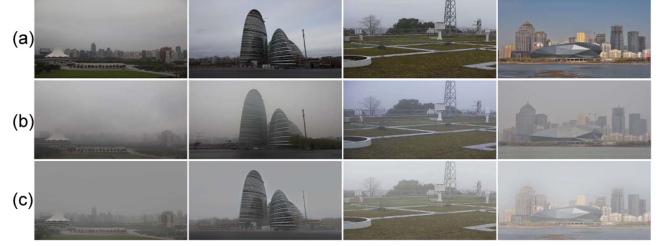


Fig. 13. (a) the reference images; (b) the real-world hazy images; (c) our synthesized hazy images. The synthetic hazy images are realistic and it is difficult to distinguish them from real-world ones with the naked eye.

TABLE I
COMPARISONS OF THE FID SCORES BETWEEN OURS, RESIDE AND FOGGY CITYSCAPES. THE LOWER THE SCORE, THE BETTER

Metric	RESIDE-ITS [4]	RESIDE-OTS [4]	Foggy Cityscapes [10]	Ours
FID	59.36	45.73	55.21	38.29

photo-realistic, and a clear image has multiple hazy images with different haze distributions and densities. In addition, our synthetic hazy images are similar to the real-world ones, which are quite difficult to discriminate by human eyes.

Second, we compare quantitatively two publicly haze benchmark datasets and our generated hazy images. Table I reports the FID [60] scores of the hazy images generated by ours, from the RESIDE [4], and the Foggy Cityscapes [10]. We achieved the lowest score, which implies that our synthetic hazy images are much closer to the real-world hazy images. The reason behind this is that the haze distributions are diverse for our synthetic haze images and have stronger correlations with real-world hazy images.

C. Comparison With SOTA Image Dehazing Methods

Visual Quality: Fig. 14 reports the real-world hazy images and their dehazed images by different image dehazing works. Note that real-world hazy images have dense, medium, and light haze density. From it, we can observe that the proposed SSID produces dehazing results with sharper structure, brighter colors, and more details. The Deep DCP [54] seems unable to handle the sky region, which causes serious color distortion. YOLY [35] has little distortion but remains a lot hazy. UCL-Dehaze [37] is weak in dehazing and has serve color distortion. Both RefinedNet [55] and USID [36] achieve desirable results, but there is an observable dark color for the results by RefinedNet, and there remain some haze in distant areas and some color distortions in local areas for the results by USID. To sum up, the proposed SSID generates comparable dehazing results and keeps well image structure and color.

Quantitative Dehazing Quality Assessment: Five specifically designed metrics DehIQA [56], FRFSIM [57], VI [58], RI [58], and FADE [59] are used to quantitative evaluate the performance of image dehazing methods. Fig. 15 reports the dehazed images from RW-HAZE, and Table II summarizes their objective assessment results of the dehazed images. From Table II, we can observe that SSID achieves the top three scores

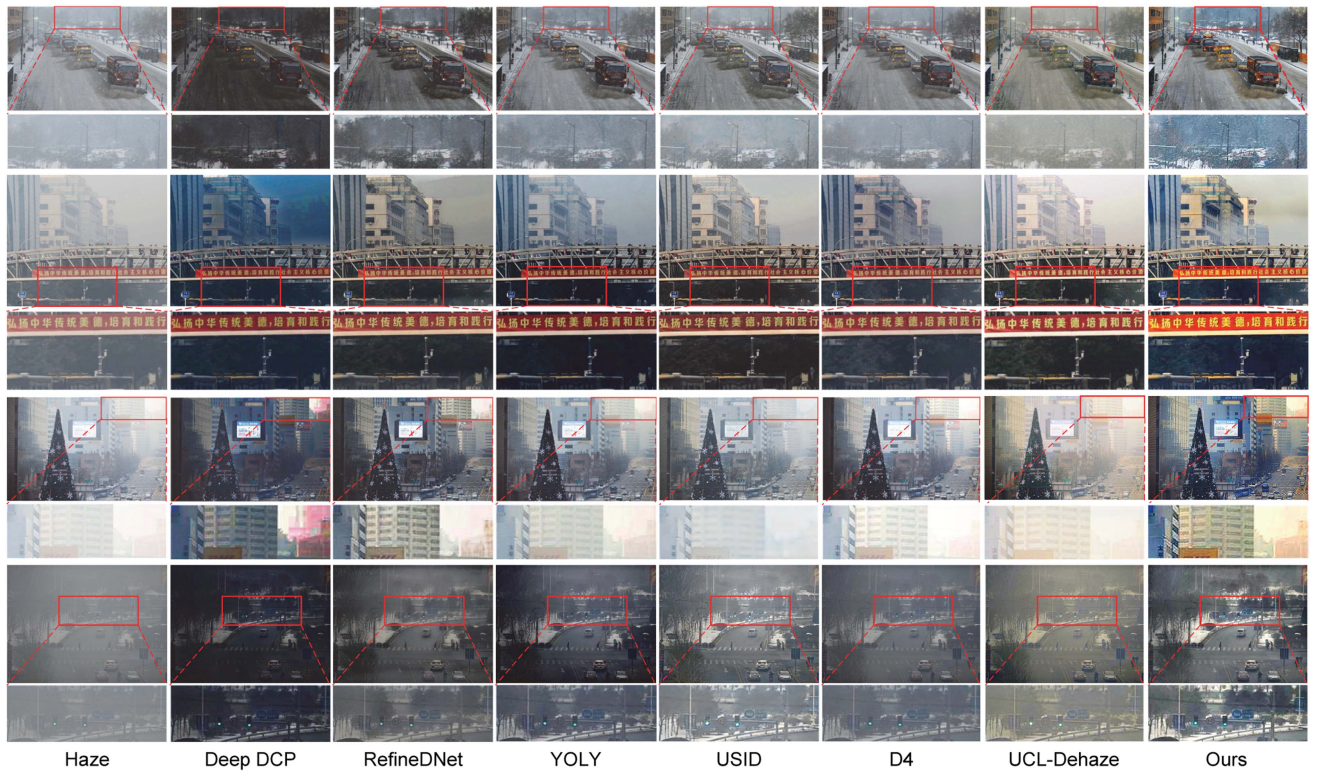


Fig. 14. Comparison of visual results on no-reference real-world hazy images of the unsupervised-based Deep DCP [54], RefineDNet [55], YOLY [35], USID [36], D4 [39], UCL-Dehaze [37] and ours, respectively.

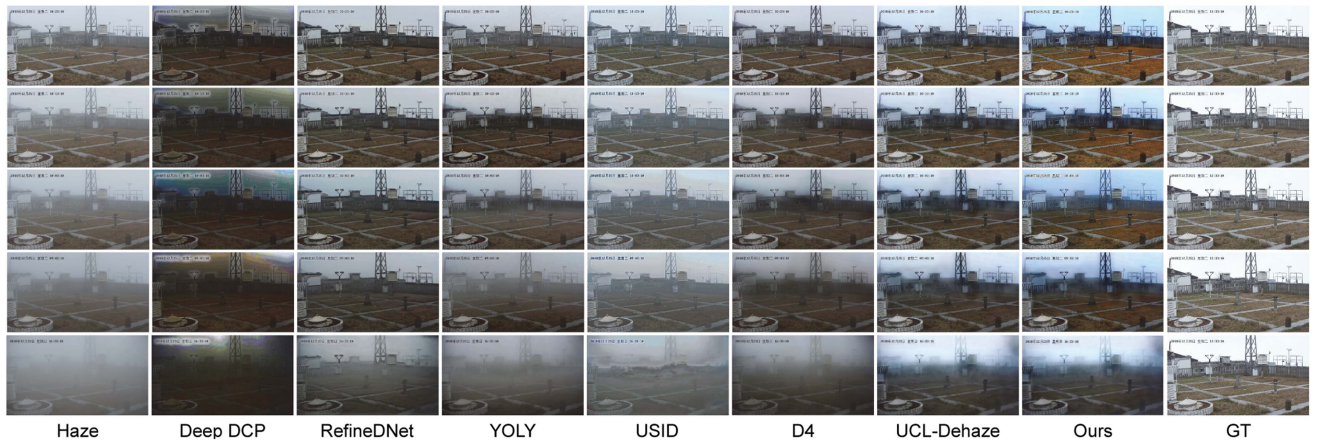


Fig. 15. Comparison of visual results on the real-world hazy images from RW-HAZE of unsupervised-based Deep DCP [54], RefineDNet [55], YOLY [35], USID [36], D4 [39], UCL-Dehaze [37] and Ours, respectively.

for all full-reference IQA metrics. Particularly, it ranks first places with DehIQA and FRFSIM, which demonstrate the powerful fitting ability on real-world hazy images. DehIQA and FRFSIM are the latest deep learning-based metrics for image dehazing assessment. RefineDNet and D4 performed best on VI and RI, respectively. They were trained using unpaired hazy images and clear ones, which provided weakly supervised labels for image dehazing. Strictly speaking, they should be considered weakly supervised learning dehazing methods. In contrast, Deep DCP only uses hazy images for training, but the dehazed images suffer from severe color distortion, it has a lot of room for

improvement. FADE is a fog-aware density estimator, and there is almost no haze residual due to severe distortion, which is the reason that Deep DCP can achieve the best results on FADE. From Fig. 15 we can see that the dehazing results of our method on real-world hazy images are more thorough while maintaining less distortion.

Furthermore, Fig. 16 shows the score of PSNR and SSIM on each dehazed image obtained from I-HAZE and O-HAZE datasets by seven dehazing methods including DCP-based and six unsupervised learning-based. From it, we can see that the dehazed image obtained by the proposed method achieves the

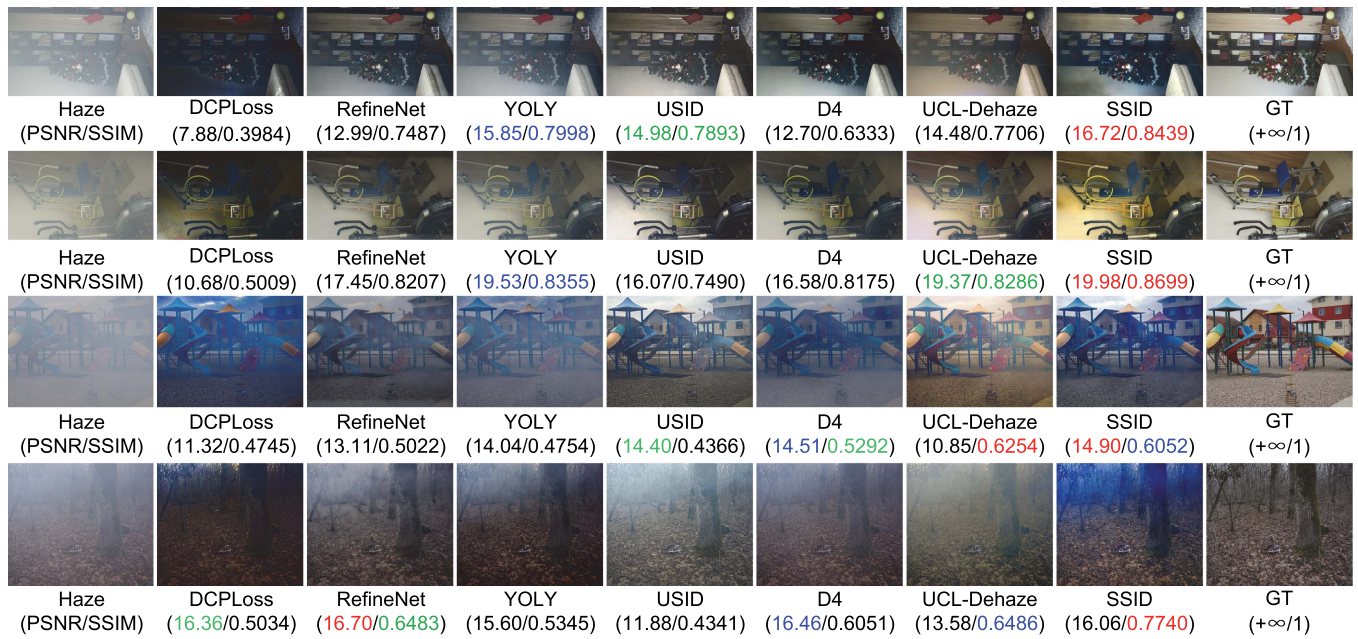


Fig. 16. Comparison of visual results on the real-world hazy image from I-haze [52] and O-haze [53]. The top three performance values are indicated with red, blue, and green, respectively.

TABLE II
QUANTITATIVE RESULTS USING FULL REFERENCE IMAGE QUALITY
ASSESSMENT METRICS OF DIFFERENT HAZE LEVELS

Method	DehIQA↓	FRFSIM↑	VI↑	RI↑	FADE↓
Deep DCP [54]	0.5023	0.1274	0.7588	0.9380	0.8282
RefineDNet [55]	0.4308	0.2392	0.8891	0.9606	1.4584
YOLY [35]	0.4501	0.2341	0.8476	0.9625	1.9294
USID [36]	0.4472	0.3581	0.8377	0.9582	2.1455
D4 [39]	0.4351	0.2829	0.8665	0.9641	1.7718
UCL-Dehaze [37]	0.4490	0.3685	0.8356	0.9738	1.4339
SSID (Ours)	0.3913	0.4138	0.8887	0.9623	1.4889

The top two performance values are indicated with red, blue, respectively.

top three scores on PSNR and SSIM, with almost no residual haze but slight color distortion.

D. Effectiveness of the Image Hazing Pipeline

First, to prove that our proposed image hazing pipeline for paired data generation can boost the performance of haze removal, we retrain our SSID on a widely-used synthetic hazy image dataset RESIDE-OTS [4], where the input paired images are augmented with random flipping and random rotation for data augmentation. Finally, the models are tested on the same real-world hazy images. The number of training data remained the same when we retrained SSID. The qualitative comparison results are illustrated in Fig. 17. We can see that the dehazed images generated by the model trained on the REDISE-OTS remain more haze and struggle in color recovery. Tabel III presents the five image dehazing quality assessment metrics results. From it, we can conclude that the model trained on the images generated by our redefined hazy image generator is both superior and robust. This demonstrates that the hazy images synthesized with our proposed generator are more effective than those



Fig. 17. Ablation results of the proposed image hazing pipeline. We retrain our dehazing model on RESIDE-OTS with the same number of training data to verify the effectiveness of the pseudo labels generated by our hazy image generation pipeline.

TABLE III
COMPARISON WITH THE TRADITIONAL DATA AUGMENTATION STRATEGIES

Method	DehIQA↓	FRFSIM↑	VI↑	RI↑	FADE↓
SSID (RESIDE-OTS)	0.4351	0.3735	0.8459	0.9248	1.6810
SSID (Our Pseudo Labels)	0.3913	0.4138	0.8887	0.9623	1.4889

produced by traditional augmentation strategies. The reason may be that although traditional enhancement strategies can increase the amount of training data and help reduce model overfitting, these enhanced data cannot always be guaranteed to be beneficial. For example, random rotation may shift certain regions out of the picture, leaving those regions filled with zeros, which can make them meaningless for image dehazing and introduce additional interference during model training. Additionally, traditional data augmentation strategies cannot enhance the diversity of haze distribution itself. In contrast, the hazy images synthesized by our redefined haze image generator not only increase

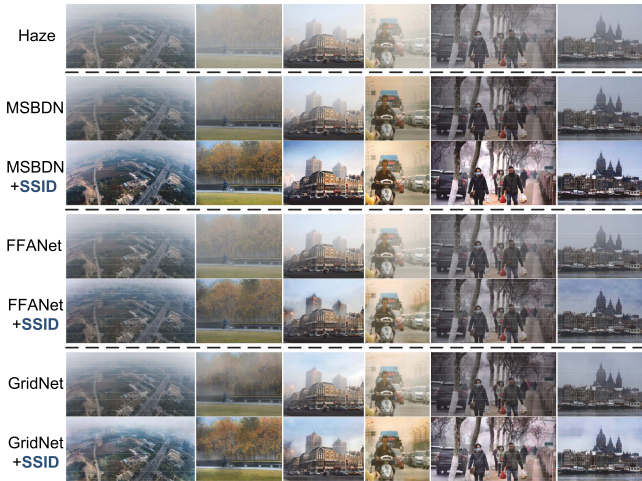


Fig. 18. Comparison of dehazing results on real-world hazy images from MSBDN [61], FFANet [62] and GridNet [24]. MSBDN represents dehazed images generated by the open-sourced model, while MSBDN+SSID represents dehazed images generated by the model trained with our self-supervised framework.

TABLE IV
COMPARISON OF DEHAZING RESULTS ON REAL-WORLD HAZY IMAGES

Method	PSNR $\uparrow$	SSIM $\uparrow$
MSBDN	11.91	0.4786
MSBDN+SSID	13.41	0.5824
Gain	+1.50	+0.1038
FFANet	12.84	0.4528
FFANet+SSID	13.63	0.4899
Gain	+0.79	+0.0371
GridNet	13.13	0.6020
GridNet+SSID	16.29	0.6688
Gain	+3.16	+0.0668

the amount of training data, but also improve the diversity of haze distribution while ensuring the quality of the images. Furthermore, our proposed hazy image generator only inputs clear images without extra manual labels, which is essential for developing our self-supervised image dehazing framework. However, traditional data augmentation cannot achieve this.

Second, to demonstrate that the proposed framework can significantly improve the performance of existing dehazing methods on real-world hazy images, two supervised learning-based dehazing models, namely MSBDN [61] and GridNet [24], are directly integrated into the framework, where only clear images are fed into the models for training in a self-supervised manner, and all model hyper-parameters remain the same as when published. Finally, the dehazing results are compared with the open-sourced models after supervised training by the authors. Fig. 18 reports the experimental results. Compared with the native models, SSID greatly improves their generalization capabilities and obtains much better-dehazed images. In addition, Table IV compares the quantitative PSNR/SSIM results before and after the native models are trained under the proposed SSID framework, we can see that both MSBDN and GridNet achieve higher scores in terms of PSNR/SSIM when they are trained under the proposed self-supervised framework, which also proves

TABLE V
AVERAGE RUNNING TIME, PARAMETERS AND FLOPS

Method	Runtime(s)	Parameters(M)	FLOPs
Deep DCP [54]	0.6532	0.507	$1.30e^{11}$
RefineDNet [55]	0.4673	65.79	$3.00e^{11}$
YOLY [35]	55.333	430.6	$1.05e^{10}$
USID [36]	0.0543	3.770	$1.60e^{11}$
D4 [39]	0.0092	9.518	$8.81e^9$
UCL-Dehaze [37]	0.0428	14.117	$7.88e^{11}$
SSID (Ours)	0.0090	2.779	$2.32e^{10}$

that SSID can promote the generalization capability of deep dehazing models. This also demonstrates that when model homogeneity is serious, high-quality data becomes more and more important.

E. Visualization in Latent Space:

To demonstrate the performance of our proposed image dehazing model, we visualize the dehazed images in the latent space. Specifically, we utilized the existing classification network ResNet101 [47] to extract the latent space features of our dehazed, clear, and hazy images, respectively. A total of 1,000 images were randomly selected from each category. We then applied the widely used latent space visualization algorithm t-distributed stochastic neighbor embedding [63], to visualize the latent space of these three categories of images within the ResNet101 network. Fig. 19 shows the visualization of the images in the latent space. We can see that there is a significant difference in distribution between our dehazed images and the hazy images in the latent space (Fig. 19(b)), which is similar to the distribution difference between the clear images and the hazy ones (Fig. 19(c)). In addition, Fig. 19(a) shows that it is difficult to distinguish the distribution difference between the dehazed images and the clear images. These results indirectly prove that our proposed model has excellent dehazing performance and its generated dehazed images have superior realism.

F. Running Time and Model Complexity:

We evaluate the run time and model complexity with the package of 'thop' in Python. The average running time of each method was computed for 500 hazy images of 512×512 pixels. From Table V, the proposed SSID ranks first in terms of the running time. The parameters of the network are only 2.779 M, which is also the smallest. Some existing models have lower parameters or FLOPs, but longer runtime.

G. Ablation Study:

To demonstrate the effectiveness of the proposed multi-directional haze distribution synthesis strategy, we set two typical synthetic haze distributions, horizontal and vertical, as the baseline. Table VI reports the quantitative dehazing results of the real-world hazy images. All evaluation metrics are positively improved compared to the baseline, which demonstrates that which demonstrates that our proposed multi directional haze distribution synthesis strategy is effective and can help to improve the dehazing capability of SSID.

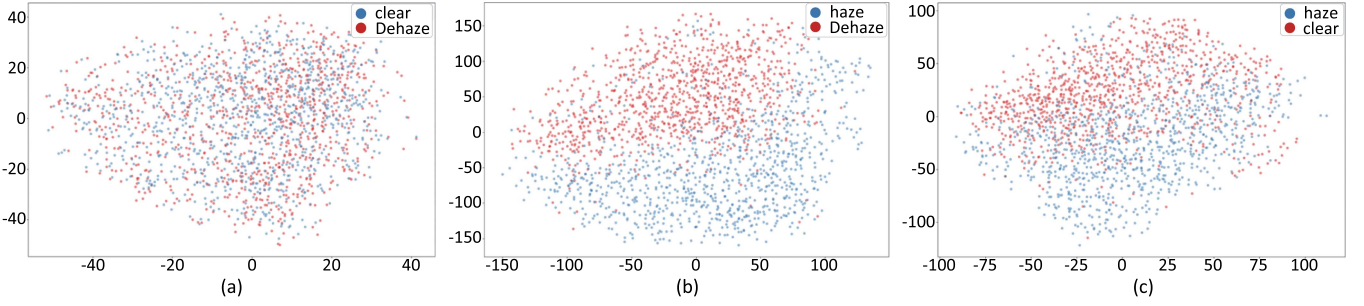


Fig. 19. Illustrate the latent space comparison between our dehazed, clear, and hazy images. (a) is the distribution of clear images and our dehazed images in latent space. (b) is the distribution of hazy images and our dehazed images in latent space. (c) is the distribution of clear images and our hazy images in latent space.

TABLE VI
ABALTON STUDY OF THE PROPOSED MULTIPLE HAZE DISTRIBUTION

Method	DehQA↓	FRFSIM↑	VI↑	RI↑	FADE↓
Baseline	0.4261	0.3892	0.8721	0.9521	1.5620
SSID	0.3913	0.4138	0.8887	0.9623	1.4889
Gain	+0.0348	+0.0246	+0.0166	+0.0102	+0.0731

V. LIMITATION

In this work, we have provided a detailed analysis of the motivation behind the proposed hazy image generator by customizing the transmission maps and redefining the transmission function for achieving SSID, and a series of experiments have been done to prove its effectiveness. Nevertheless, the theoretical derivation might be still not deep and sufficient enough. For future work, we will further optimize the existing atmospheric scattering model and combine it with a generative large model to simulate real hazy images, thereby promoting the robustness of the dehazing model in handling a wide range of real-world scenarios.

VI. CONCLUSION

To facilitate the training of deep image dehazing models and alleviate the domain shift issue, this work synthetic diverse hazy images by proposing an unsupervised hazy image generator via redefining the transmission function. In addition, based on the hazy image generator, we construct a self-supervised image dehazing framework on only leverages the observed clear image itself, without using human-annotated labels. The extensive experimental results on real-world hazy images prove that the proposed image dehazing framework achieves the best performance for unsupervised learning-based image dehazing. In turn, it also demonstrates that the proposed hazy image generator is effective and synthesizes more realistic hazy images, and implicitly proves that current synthetic hazy image datasets can not fully meet the requirements for training existing dehazing models. For future work, we will further improve the training datasets by collecting more real-world well-paired clear and improving the model's ability to address domain shift issue.

REFERENCES

- [1] C. Sakaridis, D. Dai, S. Hecker, and L. Van Gool, "Model adaptation with synthetic and real data for semantic dense foggy scene understanding," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 687–704.
- [2] Y. J. Li et al., "Cross-domain adaptive teacher for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 7581–7590.
- [3] E. McCartney, *Optics of the Atmosphere: Scattering by Molecules and Particles*. New York, NY, USA: Wiley, p. 421, 1976.
- [4] B. Li et al., "Benchmarking single-image dehazing and beyond," *IEEE Trans. Image Process.*, vol. 28, no. 1, pp. 492–505, Jan. 2019.
- [5] C. Liu et al., "Non-homogeneous haze data synthesis based real-world image dehazing with enhancement-and-restoration fused CNNs," *Comput. Graph.*, vol. 106, pp. 45–57, 2022.
- [6] J. Chen, G. Yang, M. Xia, and D. Zhang, "From depth-aware haze generation to real-world haze removal," *Neural Comput. Appl.*, vol. 35, no. 11, pp. 8281–8293, 2023.
- [7] B. Li et al., "Unsupervised neural rendering for image hazing," *IEEE Trans. Image Process.*, vol. 31, pp. 3987–3996, 2022.
- [8] C. Ancuti, C. O. Ancuti, and C. De Vleeschouwer, "D-HAZY: A dataset to evaluate quantitatively dehazing algorithms," in *Proc. 2017 IEEE Int. Conf. Image Process.*, 2016, pp. 2226–2230.
- [9] Y. Zhang, L. Ding, and G. Sharma, "Hazerd: An outdoor scene dataset and benchmark for single image dehazing," in *Proc. 2017 IEEE Int. Conf. Image Process.*, 2017, pp. 3205–3209.
- [10] C. Sakaridis, D. Dai, and L. V. Gool, "Semantic foggy scene understanding with synthetic data," *Int. J. Comput. Vis.*, vol. 126, no. 9, pp. 973–992, 2018.
- [11] C. Zhang et al., "Density-aware haze image synthesis by self-supervised content-style disentanglement," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 7, pp. 4552–4572, Jul. 2022.
- [12] Y. Li et al., "Image hazing and dehazing: From the viewpoint of two-way image translation with a weakly supervised framework," *IEEE Trans. Multimedia*, vol. 25, pp. 4704–4717, 2023.
- [13] L. Li et al., "Semi-supervised image dehazing," *IEEE Trans. Image Process.*, vol. 29, pp. 2766–2779, 2020.
- [14] Y. Liu et al., "From synthetic to real: Image dehazing collaborating with unlabeled real data," in *Proc. 29th ACM Int. Conf. Multimedia*, 2021, pp. 50–58.
- [15] Y. Shao, L. Li, W. Ren, C. Gao, and N. Sang, "Domain adaptation for image dehazing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 2808–2817.
- [16] J. P. Tarel, N. Hautiere, A. Cord, D. Gruyer, and H. Halmaoui, "Improved visibility of road scene images under heterogeneous fog," in *Proc. 2010 IEEE Intell. Veh. Symp.*, 2010, pp. 478–485.
- [17] J. Tarel et al., "Vision enhancement in homogeneous and heterogeneous fog," *IEEE Intell. Transp. Syst. Mag.*, vol. 4, no. 2, pp. 6–20, Feb. 2012.
- [18] R. Q. Wu, Z. P. Duan, C. L. Guo, Z. Chai, and C. Li, "RidCP: Revitalizing real image dehazing via high-quality codebook priors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 22282–22291.
- [19] T. C. Wang et al., "High-resolution image synthesis and semantic manipulation with conditional GANs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 8798–8807.
- [20] Y. Xie, H. Wei, Z. Liu, X. Wang, and X. Ji, "SynFog: A photo-realistic synthetic fog dataset based on end-to-end imaging simulation for advancing real-world defogging in autonomous driving," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 21763–21772.
- [21] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2341–2353, Dec. 2010.
- [22] Q. Zhu, J. Mai, and L. Shao, "A fast single image haze removal algorithm using color attenuation prior," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3522–3533, Nov. 2015.

- [23] W. Ren et al., "Gated fusion network for single image dehazing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 3253–3261.
- [24] X. Liu, Y. Ma, Z. Shi, and J. Chen, "GridDehazeNet: Attention-based multi-scale network for image dehazing," in *Proc. 2019 IEEE Int. Conf. Comput. Vis.*, 2019, pp. 7313–7322.
- [25] X. Song et al., "WSAMF-Net: Wavelet spatial attention-based multistream feedback network for single image dehazing," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 2, pp. 575–588, Feb. 2023.
- [26] G. Zhang, W. Fang, Y. Zheng, and R. Wang, "SDBAD-Net: A spatial dual-branch attention dehazing network based on meta-former paradigm," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 1, pp. 60–70, Jan. 2024.
- [27] Y. Feng, X. Meng, F. Zhou, W. Lin, and Z. Su, "Real-world non-homogeneous haze removal by sliding self-attention wavelet network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 10, pp. 5470–5485, Oct. 2023.
- [28] Z. Chen, Z. He, and Z.-M. Lu, "DEA-Net: Single image dehazing based on detail-enhanced convolution and content-guided attention," *IEEE Trans. Image Process.*, vol. 33, pp. 1002–1015, 2024.
- [29] Y. Dong, Y. Liu, H. Zhang, S. Chen, and Y. Qiao, "FD-GAN: Generative adversarial networks with fusion-discriminator for single image dehazing," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, no. 07, 2020, pp. 10729–10736.
- [30] Z. Chen, Y. Wang, Y. Yang, and D. Liu, "PSD: Principled synthetic-to-real dehazing guided by physical priors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 7180–7189.
- [31] T. Jia, J. Li, L. Zhuo, and T. Yu, "Semi-supervised single-image dehazing network via disentangled meta-knowledge," *IEEE Trans. Multimedia*, vol. 26, pp. 2634–2647, 2024.
- [32] S. Li, Y. Zhou, W. Ren, and W. Xiang, "Pfonet: A progressive feedback optimization network for lightweight single image dehazing," *IEEE Trans. Image Process.*, vol. 32, pp. 6558–6569, 2023.
- [33] J. Gui et al., "A comprehensive survey and taxonomy on single image dehazing based on deep learning," *ACM Comput. Surv.*, vol. 55, no. 13s, pp. 1–37, 2023.
- [34] B. Li et al., "Zero-shot image dehazing," *IEEE Trans. Image Process.*, vol. 29, pp. 8457–8466, 2020.
- [35] B. Li et al., "You only look yourself: Unsupervised and untrained single image dehazing neural network," *Int. J. Comput. Vis.*, vol. 129, no. 5, pp. 1754–1767, 2021.
- [36] J. Li, Y. Li, L. Zhuo, L. Kuang, and T. Yu, "USID-Net: Unsupervised single image dehazing network via disentangled representations," *IEEE Trans. Multimedia*, vol. 25, pp. 3587–3601, 2023.
- [37] Y. Wang et al., "Ucl-dehaze: Toward real-world image dehazing via unsupervised contrastive learning," *IEEE Trans. Image Process.*, vol. 33, pp. 1361–1374, 2024.
- [38] J. Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. 2017 IEEE Int. Conf. Comput. Vis.*, 2017, pp. 2223–2232.
- [39] Y. Yang et al., "Self-augmented unpaired image dehazing via density and depth decomposition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 2037–2046.
- [40] Y. Liang, B. Wang, W. Zuo, J. Liu, and W. Ren, "Self-supervised learning and adaptation for single image dehazing," in *Proc. IJCAI*, 2022, pp. 1137–1143.
- [41] L. Jing and Y. Tian, "Self-supervised visual feature learning with deep neural networks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 11, pp. 4037–4058, Nov. 2021.
- [42] K. Nguyen and S. Todorovic, "A self-supervised GAN for unsupervised few-shot object recognition," in *Proc. 25th Int. Conf. Pattern Recognit.*, 2021, pp. 3225–3231.
- [43] G. Baykal, F. Ozelik, and G. Unal, "Exploring DeshuffleGANs in self-supervised generative adversarial networks," *Pattern Recognit.*, vol. 122, 2022, Art. no. 108244.
- [44] G. Tkachev, S. Frey, and T. Ertl, "S4: Self-supervised learning of spatiotemporal similarity," *IEEE Trans. Vis. Comput. Graph.*, vol. 28, no. 12, pp. 4713–4727, Dec. 2022.
- [45] Z. Li and N. Snavely, "Megadepth: Learning single-view depth prediction from internet photos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 2041–2050.
- [46] C. D. Lai, M. Xie, and D. N. P. Murthy, "A modified weibull distribution," *IEEE Trans. Rel.*, vol. 52, no. 1, pp. 33–37, Jan. 2004.
- [47] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [48] Q. Hou, D. Zhou, and J. Feng, "Coordinate attention for efficient mobile network design," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 13708–13717.
- [49] D. Lin, C. Lu, H. Huang, and J. Jia, "RSCM: Region selection and concurrency model for multi-class weather recognition," *IEEE Trans. Image Process.*, vol. 26, no. 9, pp. 4154–4167, Sep. 2017.
- [50] C. Lu, D. Lin, J. Jia, and C. Tang, "Two-class weather classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, 2014, pp. 3718–3725.
- [51] J. Chen, S. Wang, X. Liu, and G. Yang, "RW-HAZE: A real-world benchmark dataset to evaluate quantitatively dehazing algorithms," in *Proc. 2022 IEEE Int. Conf. Image Process.*, 2022, pp. 11–15.
- [52] C. Ancuti, C. Ancuti, R. Timofte, and C. D. Vleeschouwer, "I-HAZE: A dehazing benchmark with real hazy and haze-free indoor images," in *Proc. Adv. Concepts Intell. Vis. Syst.*, 2018, pp. 620–631.
- [53] C. Ancuti, C. Ancuti, R. Timofte, and C. D. Vleeschouwer, "O-HAZE: A dehazing benchmark with real hazy and haze-free outdoor images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, 2018, pp. 754–762.
- [54] A. Golts, D. Freedman, and M. Elad, "Unsupervised single image dehazing using dark channel prior loss," *IEEE Trans. Image Process.*, vol. 29, pp. 2692–2701, 2020.
- [55] S. Zhao, L. Zhang, Y. Shen, and Y. Zhou, "Refinednet: A weakly supervised refinement framework for single image dehazing," *IEEE Trans. Image Process.*, vol. 30, pp. 3391–3404, 2021.
- [56] J. Chen, G. Yang, S. Wang, D. Wang, and X. Liao, "Image dehazing assessment: A real-world dataset and a haze density-aware criteria," *IEEE Trans. Multimedia*, vol. 26, pp. 6037–6049, 2024.
- [57] W. Liu, F. Zhou, T. Lu, J. Duan, and G. Qiu, "Image defogging quality assessment: Real-world database and method," *IEEE Trans. Image Process.*, vol. 30, pp. 176–190, 2020.
- [58] S. Zhao, L. Zhang, S. Huang, Y. Shen, and S. Zhao, "Dehazing evaluation: Real-world benchmark datasets, criteria, and baselines," *IEEE Trans. Image Process.*, vol. 29, pp. 6947–6962, 2020.
- [59] L. K. Choi, J. You, and A. C. Bovik, "Referenceless prediction of perceptual fog density and perceptual image defogging," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3888–3901, Nov. 2015.
- [60] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local NASH equilibrium," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 6626–6637.
- [61] H. Dong et al., "Multi-scale boosted dehazing network with dense feature fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 2157–2167.
- [62] X. Qin, Z. Wang, Y. Bai, X. Xie, and H. Jia, "FFA-Net: Feature fusion attention network for single image dehazing," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 07, pp. 11908–11915.
- [63] G. E. Hinton and S. Roweis, "Stochastic neighbor embedding," in *Proc. Adv. Neural Inf. Process. Syst.*, 2002, vol. 15, pp. 857–864.